Course Title : HIGH PERFORMANCE COMPUTING Lab Ass-5

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Lab: Loop Scheduling Comparison
Task 1: Code (reuse Week-3 code, only change directive)

```
#pragma omp parallel for schedule(runtime)
reduction(+:sum)
for (long i = 0; i &lt; N; i++)
{ sum += i;
}
```

Run commands:

```
gcc -fopenmp sum_openmp.c -o sum_openmp

export OMP_SCHEDULE=static
export OMP_NUM_THREADS=8
./sum_openmp

export OMP_SCHEDULE=dynamic
```

04-02-2026 EOD

```
./sum_openmp

export OMP_SCHEDULE=guided
./sum_openmp
```

Fill the table below:

Scheduling	Time	Observation
Static	?	?
Dynamic	?	?
Guided	?	?

```
  GNU nano 7.2                                          sum_serial.c
## Task 1: Loop Scheduling Comparison (schedule(runtime))

```c
#include <stdio.h>
#include <omp.h>

int main() {
    long N = 100000000;
    long long sum = 0;

    double start = omp_get_wtime();

    #pragma omp parallel for schedule(runtime) reduction(+:sum)
    for (long i = 0; i < N; i++) {
        sum += i;
    }

    double end = omp_get_wtime();

    printf("Sum = %lld\n", sum);
    printf("Time taken = %f seconds\n", end - start);

    return 0;
}
```

                                          [Read 106 lines]
^G Help   ^O Write Out   ^W Where Is   ^K Cut   ^T Execute   ^C Location   M-U Undo   M-A Set Mark
^X Exit   ^R Read File   ^\ Replace   ^U Paste   ^J Justify   ^/ Go To Line M-E Redo   M-6 Copy
```

```
nithish@LAPTOP-A2FOJ865:/mnt/c/Users/Sreeshma$ nano sum_openmp.c
nithish@LAPTOP-A2FOJ865:/mnt/c/Users/Sreeshma$ gcc -fopenmp sum_openmp.c -o sum_openmp
nithish@LAPTOP-A2FOJ865:/mnt/c/Users/Sreeshma$ export OMP_NUM_THREADS=8

export OMP_SCHEDULE=static
./sum_openmp

export OMP_SCHEDULE=dynamic
./sum_openmp

export OMP_SCHEDULE=guided
./sum_openmp
Sum = 4999999950000000
Time taken = 0.115592 seconds
Sum = 4999999950000000
Time taken = 4.541940 seconds
Sum = 4999999950000000
Time taken = 0.055931 seconds
```

Scheduling	Time	Observation
Static	0.115592 seconds	4999999950000000
Dynamic	4.541940 seconds	4999999950000000
Guided	0.055931 seconds	4999999950000000

2

Task 2: Synchronization Overhead Comparison

Variant-1: Reduction

#pragma omp parallel for reduction(+:sum)

Variant-2: Atomic

#pragma omp parallel for

for (...) {

```
#pragma omp atomic
sum += i;
}
```

Variant-3: Critical

```
#pragma omp parallel for
```

04-02-2026 EOD

```
for (...) {
#pragma omp
critical sum += i;
}
```

Expected trend:
Reduction → fastest
Atomic → slower
Critical → slowest

Reduction:

```
  GNU nano 7.2                    reduction.c
#include <stdio.h>
#include <omp.h>

int main() {
    long N = 100000000;
    long long sum = 0;

    #pragma omp parallel for reduction(+:sum)
    for (long i = 0; i < N; i++) {
        sum += i;
    }

    printf("Reduction Sum = %lld\n", sum);
    return 0;
}

                          [ Read 15 lines ]
^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

```
nithish@LAPTOP-A2FOJ865:/mnt/c/Users/Sreeshma$ gcc -fopenmp reduction.c -o r
eduction
./reduction
Reduction Sum = 4999999950000000
```

Atomic:

```
  GNU nano 7.2                          atomic.c *
#include <stdio.h>
#include <omp.h>

int main() {
    long N = 100000000;
    long long sum = 0;

    #pragma omp parallel for
    for (long i = 0; i < N; i++) {
        #pragma omp atomic
        sum += i;
    }

    printf("Atomic Sum = %lld\n", sum);
    return 0;
}
```

```
nithish@LAPTOP-A2FOJ865:/mnt/c/Users/Sreeshma$ nano atomic.c
nithish@LAPTOP-A2FOJ865:/mnt/c/Users/Sreeshma$ gcc -fopenmp atomic.c -o atom
ic
./atomic

Atomic Sum = 4999999950000000
```

Critical:

```
  GNU nano 7.2                    critical.c *
#include <stdio.h>
#include <omp.h>

int main() {
    long N = 100000000;
    long long sum = 0;

    #pragma omp parallel for
    for (long i = 0; i < N; i++) {
        #pragma omp critical
        sum += i;
    }

    printf("Critical Sum = %lld\n", sum);
    return 0;
}
```

```
nithish@LAPTOP-A2FOJ865:/mnt/c/Users/Sreeshma$ gcc -fopenmp critical.c -o cr
itical
./critical
Critical Sum = 4999999950000000
nithish@LAPTOP-A2FOJ865:/mnt/c/Users/Sreeshma$
nithish@LAPTOP-A2FOJ865:/mnt/c/Users/Sreeshma$
```